II. Databases

Google Cloud

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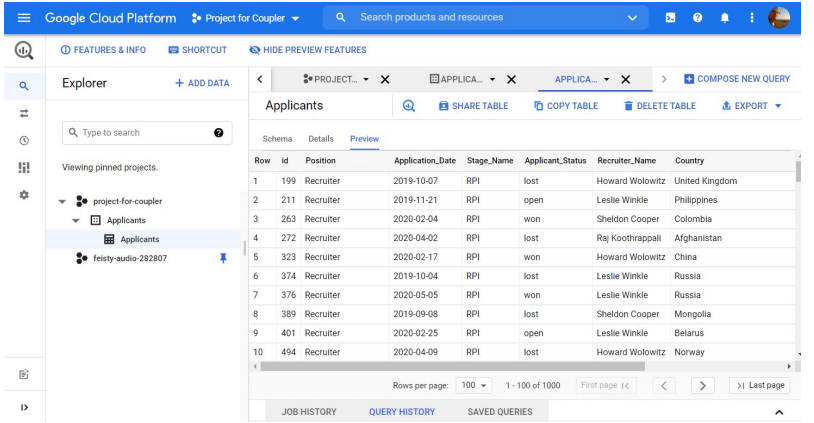
Database

Definition:
A database is a structured and organized set of information or data, stored in such a way that it can be easily accessed, manipulated, and updated.

It allows for the management and organization of large amounts of information to facilitate data access, insertion, modification, and deletion.

Databases are used in a wide range of applications, from enterprise management systems to web applications.



GCP BigQuery example

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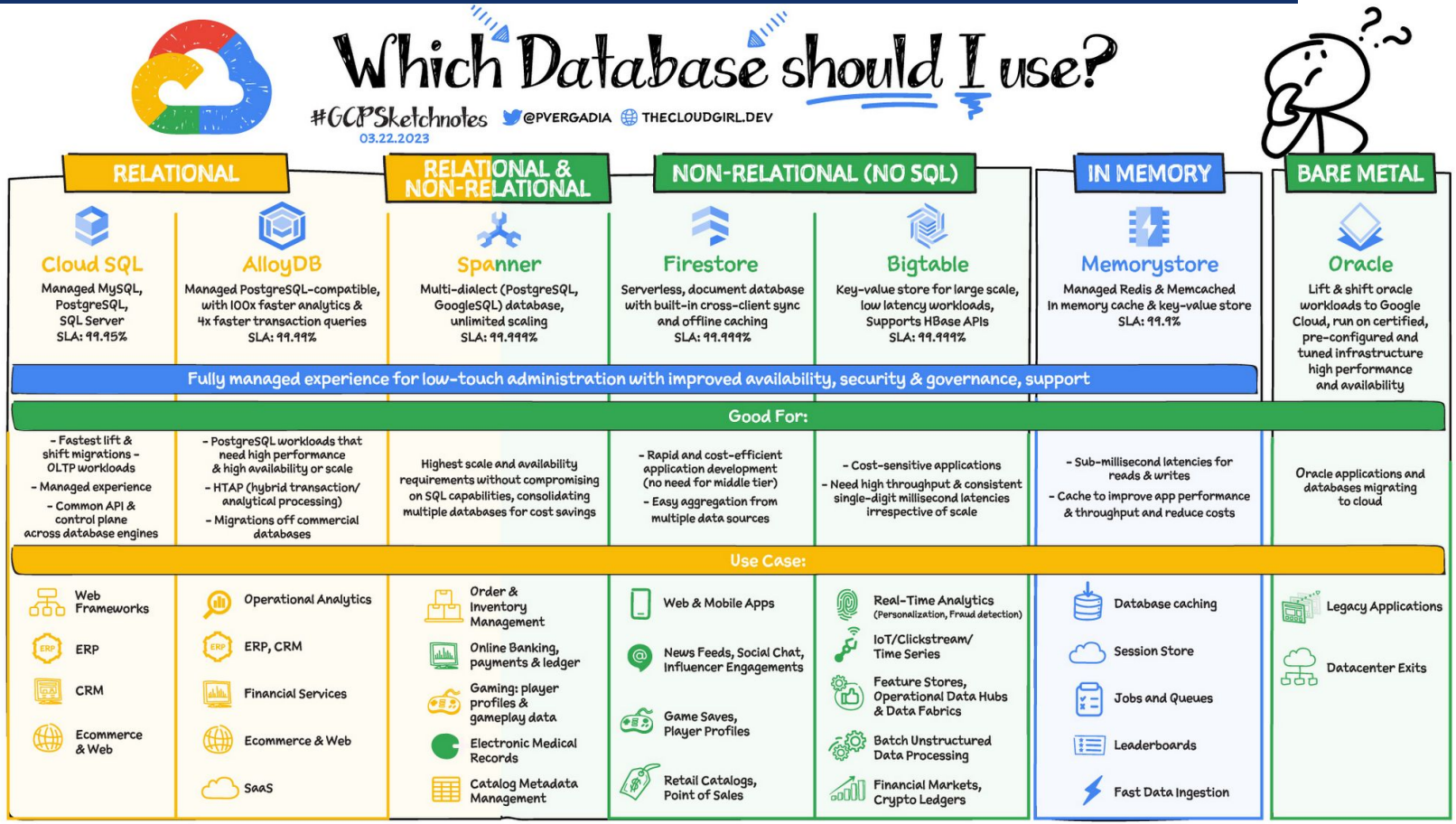

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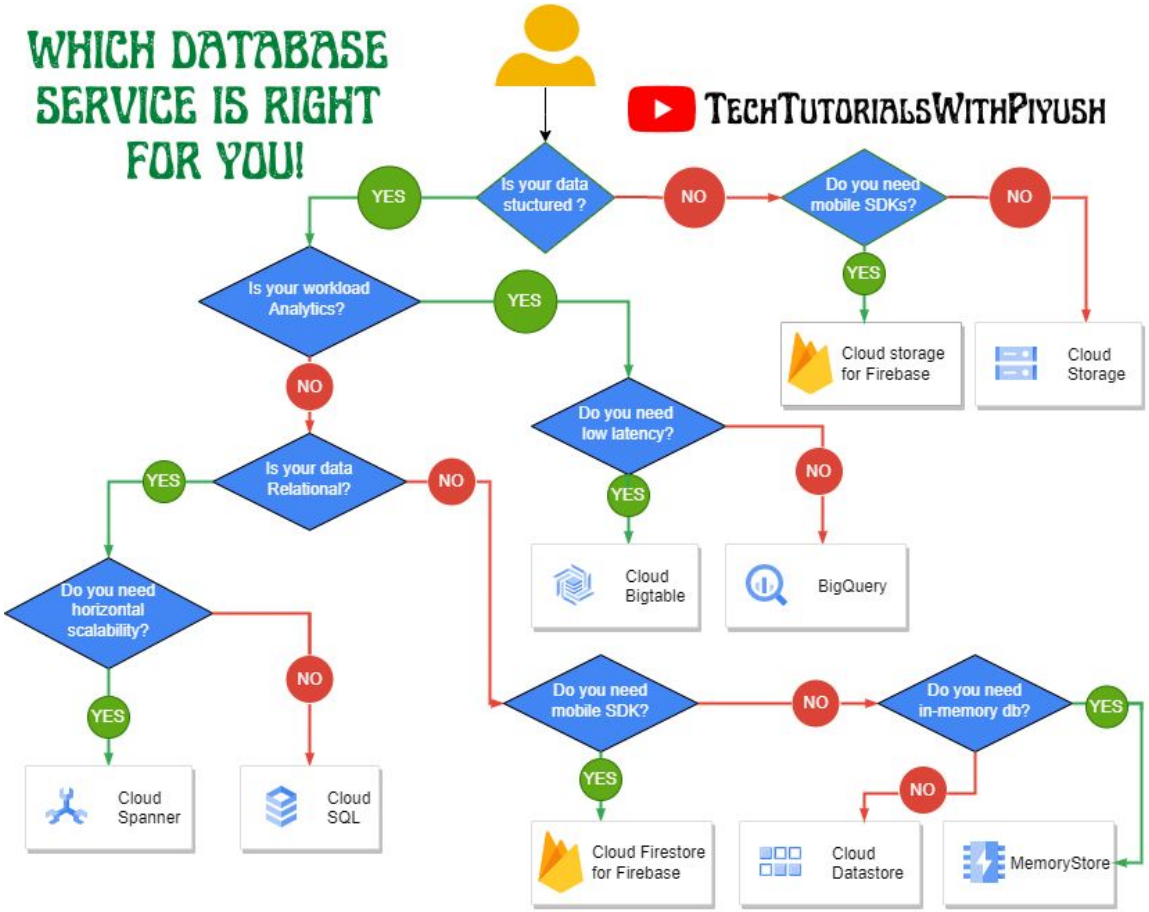

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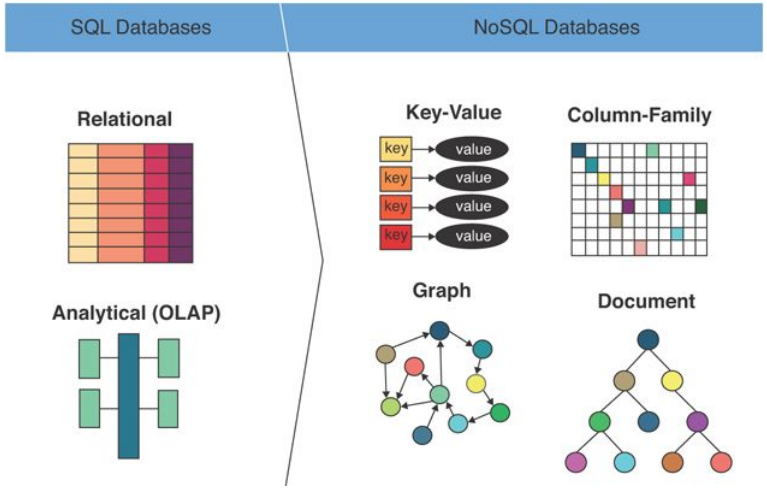
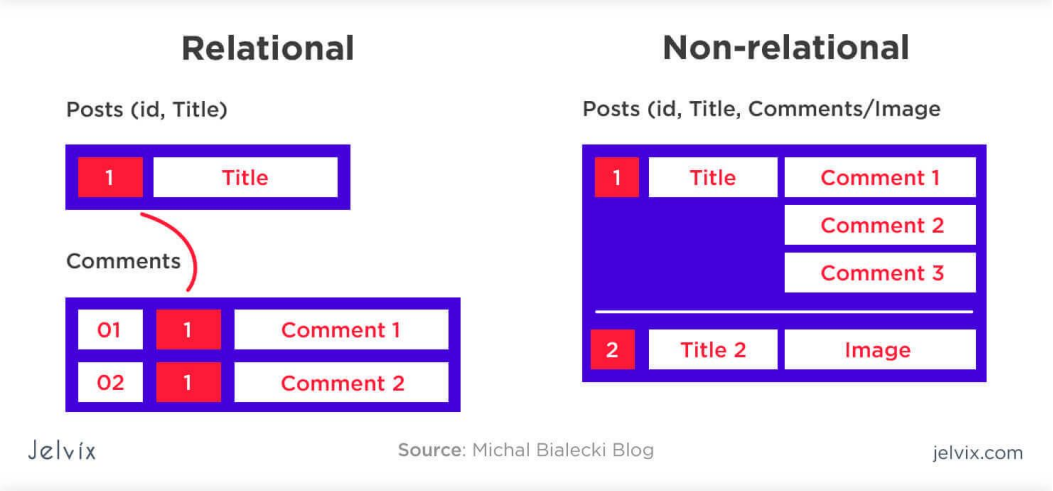
Relational VS Non-relational Database

Relational: Table like for structured data *(python array)*
Non relational: Graph or key value schema *(python dict)*





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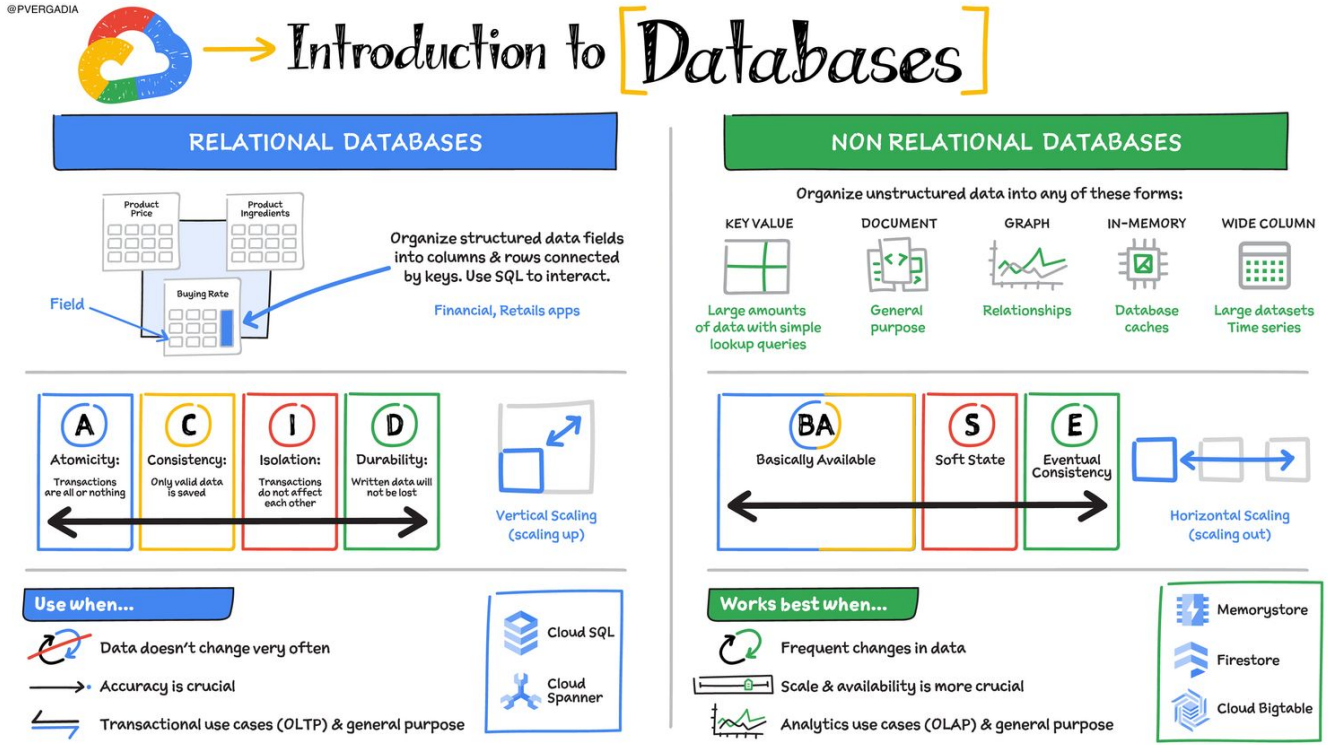
Relational VS Non-relational Database



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Relational Database

In relational databases information is stored in tables, rows and columns, which typically works best for structured data. As a result they are used for applications in which the structure of the data does not change often. SQL (Structured Query Language) is used when interacting with most relational databases. They offer ACID consistency mode for the data, which means:

Atomic: All operations in a transaction succeed or the operation is rolled back.
Consistent: On the completion of a transaction, the database is structurally sound.
Isolated: Transactions do not contend with one another. Contentious access to data is moderated by the database so that transactions appear to run sequentially.
Durable: The results of applying a transaction are permanent, even in the presence of failures.

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ACID

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OLTP vs OLAP

- Online Analytical Processing (OLAP) efficient for analytics (groupby, sum, …)
- Online Transactional Processing (OLTP) efficient for transactional (get user password, …)

Characteristic	OLTP	OLAP
Processing Type	Handles a large number of small transactions which comprises inserts, updates, deletes on the database	Handles large volumes of data with complex queries to drive business decision.
Query types	Simpler standardized queries	Complex queries
Focus	Insert, Update, Delete records from the database.	Select to aggregate data for reporting
Response time	Very fast i.e., Milliseconds	Seconds to hours, depends upon database engine
Design	Specific to Business application, e.g., e-Commerce, internet banking, booking systems etc.	Subject area specific, such as sales, Finance, HR, marketing etc.
Source	Operational data fed thru client applications.	Aggregated data from various OLTP/Data Warehouse and streaming systems
Objective	Control and run essential business operations in real time	Planning, problem solving, decision support, data discovery, reporting and insights
Data updates	Small, and quick updates initiated by client applications	Periodically data loads using schedulers, (Note: new age OLAP systems able to handle real-time data loads).
Storage requirements	Generally smaller than OLAP, as only current snapshot of the record is stored in the database.	Generally large due to aggregating large historical datasets.
Backup and recovery	Operation Data is very critical in running business operations, generally hot backups, and replications are used to meet various SLAs.	At least daily backups are needed as at the source (OLTP systems) keeps only current snapshot of the record.
Users	Customer-facing personnel, online users, system logs	Data Scientists, Business Users, Data Analysts
Data model	Normalized Data model for faster database operations	Multi-dimensional data model for query, reporting, and aggregations.

What are the key differences between OLAP and OLTP?

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Relational DB Languages

- MySQL
- PostgreSQL
- BigQuery SQL
- …

SQL allows to:

- Create, delete table or database
- Select rows under constraints
- Modify or delete rows in the table

```
-- get  employees who joined company in 2000
SELECT
   first_name
FROM
   employees
WHERE
   YEAR(hire_date) = 2000
```

SELECT clause
FROM clause
WHERE clause
Comment
Predicate

SQL syntax

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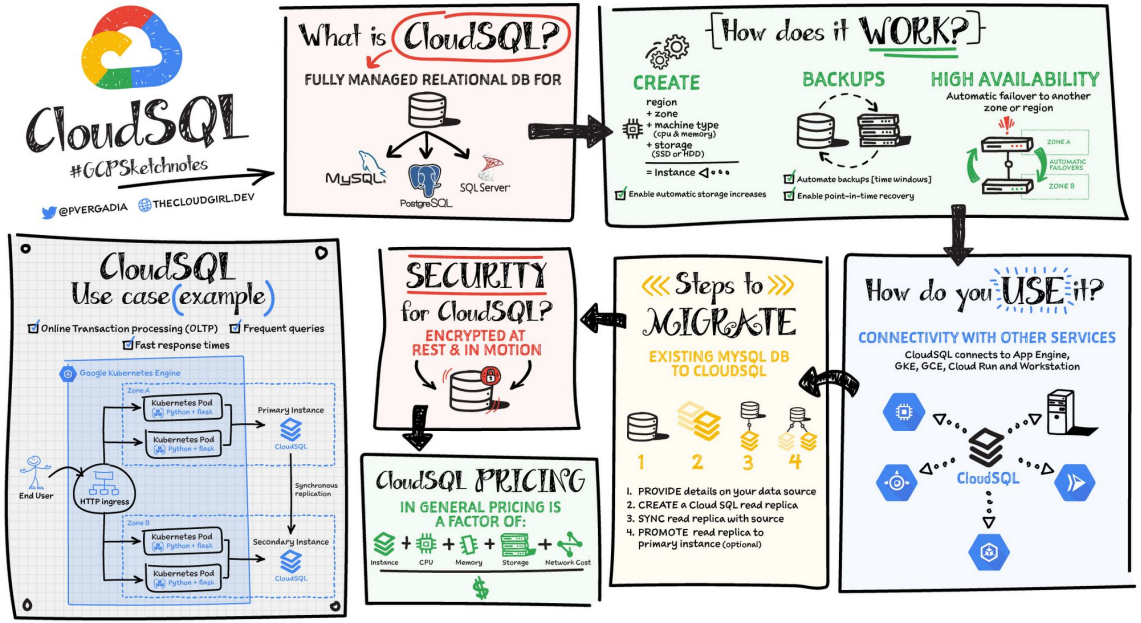
Cloud SQL - OLTP

- Relational database
- Cheapest than other GCP relational database products
- OLTP database



Cloud SQL

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Alloy DB - OLTP

Definition: AlloyDB is a fully managed, PostgreSQL-compatible database service with industry-leading performance, availability, and scale. It automates administrative tasks, such as backups, replication, patching, and capacity management, and uses adaptive algorithms and machine learning for PostgreSQL vacuum management, storage and memory management, data tiering, and analytics acceleration, so you can focus on building your applications.

- AlloyDB is up to 100x faster than standard PostgreSQL for analytical queries
- More expensive
- High availability - SLA 99.99%

AlloyDB

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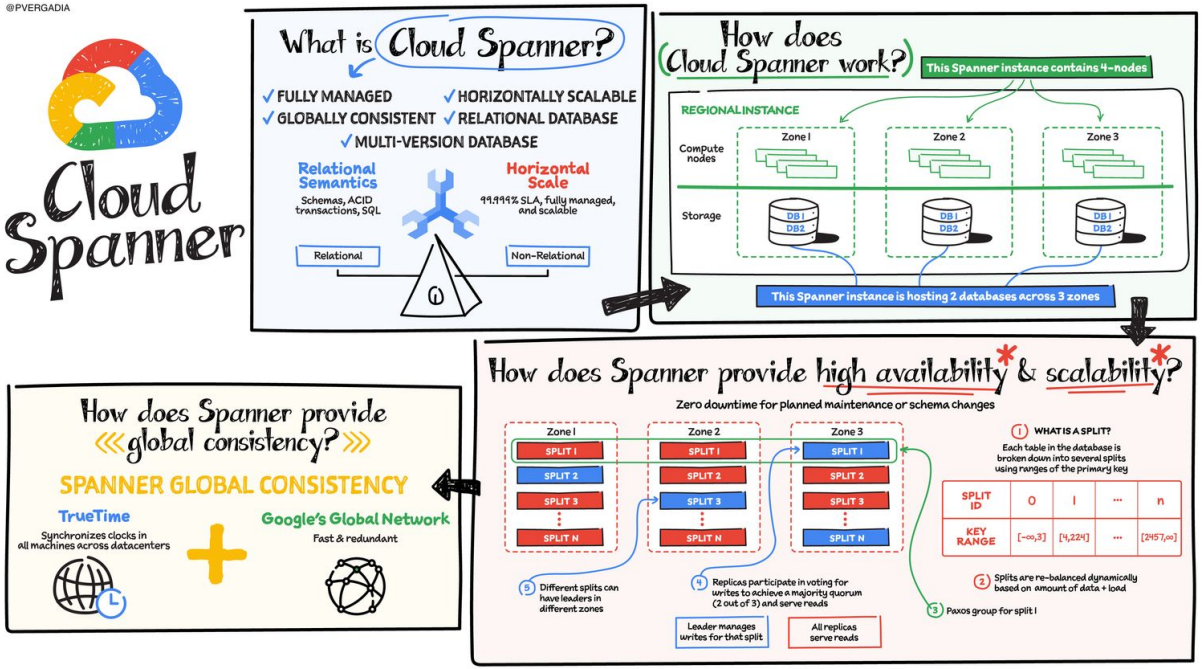
Cloud Spanner - OLTP



Spanner

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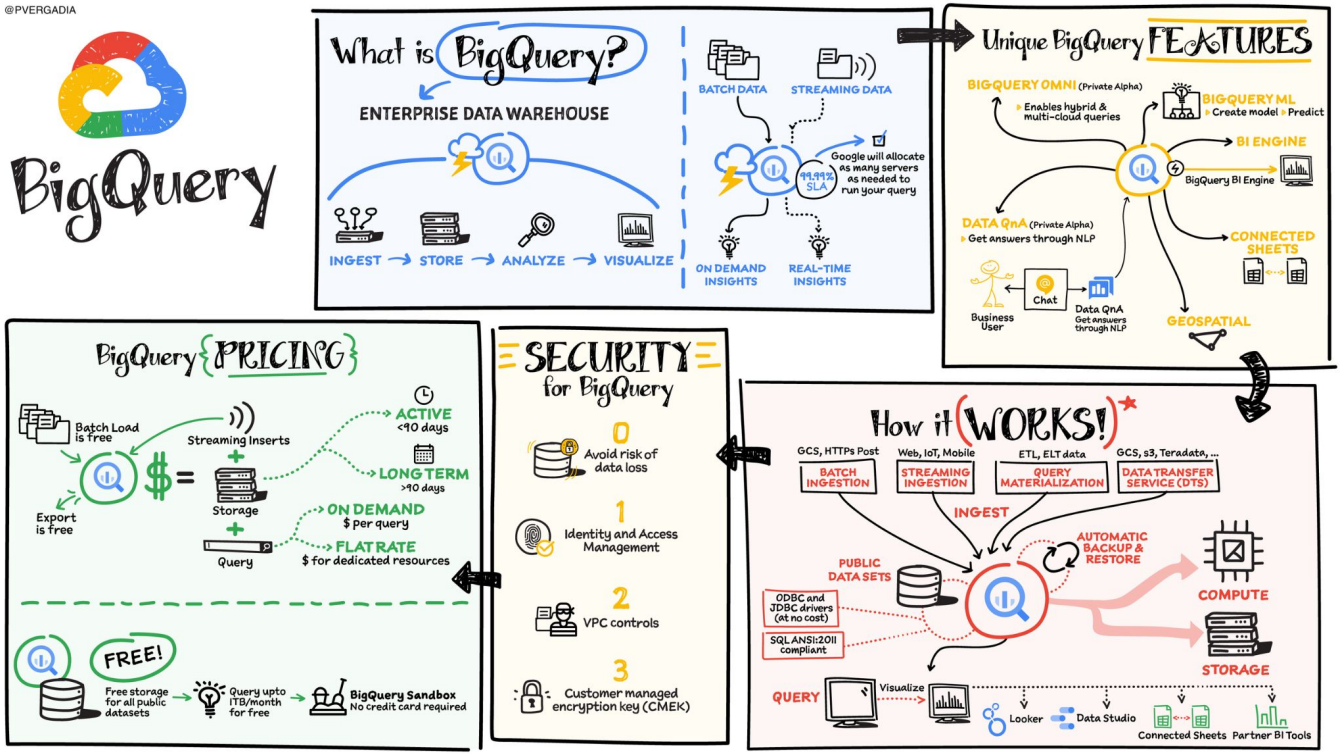
BigQuery - OLAP



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Non Relational Database

Non-relational databases (or NoSQL databases) store complex, unstructured data in a non-tabular form such as documents.

Non-relational databases are often used when large quantities of complex and diverse data need to be organized, or where the structure of the data is regularly evolving to meet new business requirements.

Unlike relational databases, they perform faster because a query doesn't have to access several tables to deliver an answer, making them ideal for storing data that may change frequently or for applications that handle many different kinds of data.

For example, an apparel store might have a database in which shirts have their own document containing all of their information, including size, brand, and color with room for adding more parameters later such as sleeve size, collars, and so on.

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Non Relational Database performance

Qualities that make NoSQL databases fast:

- Typically, they are optimized for a specific workload pattern (i.e., key-value, graph, wide-column)
- Horizontal scaling, usually using range or hashed distributions
- Eventual consistency: many NoSQL stores usually exhibit consistency at some later point (e.g., lazily at read time). However, Firestore uniquely offers strong global consistency.
- Transactions: a majority of NoSQL stores don't support cross shard transactions or flexible isolation modes. However, Firestore uniquely offers ACID transactions across shards with serializable isolation.

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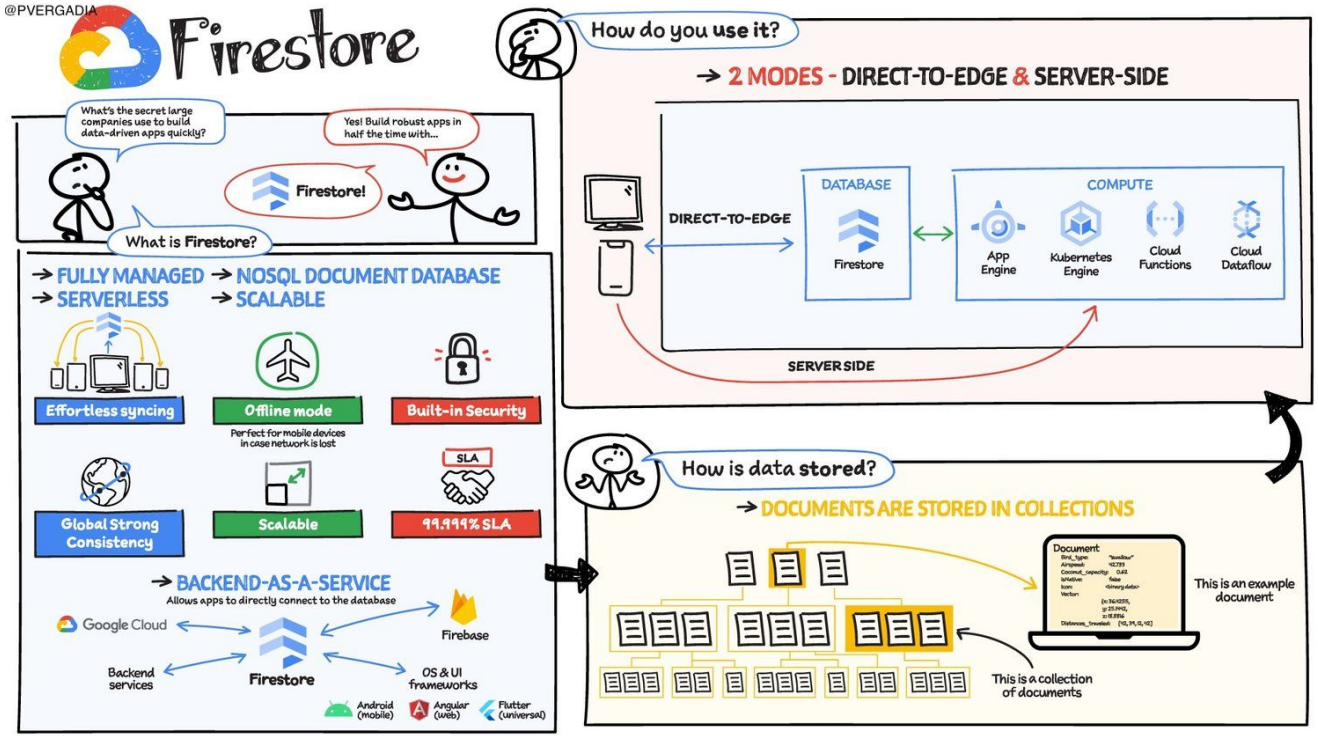
Firestore



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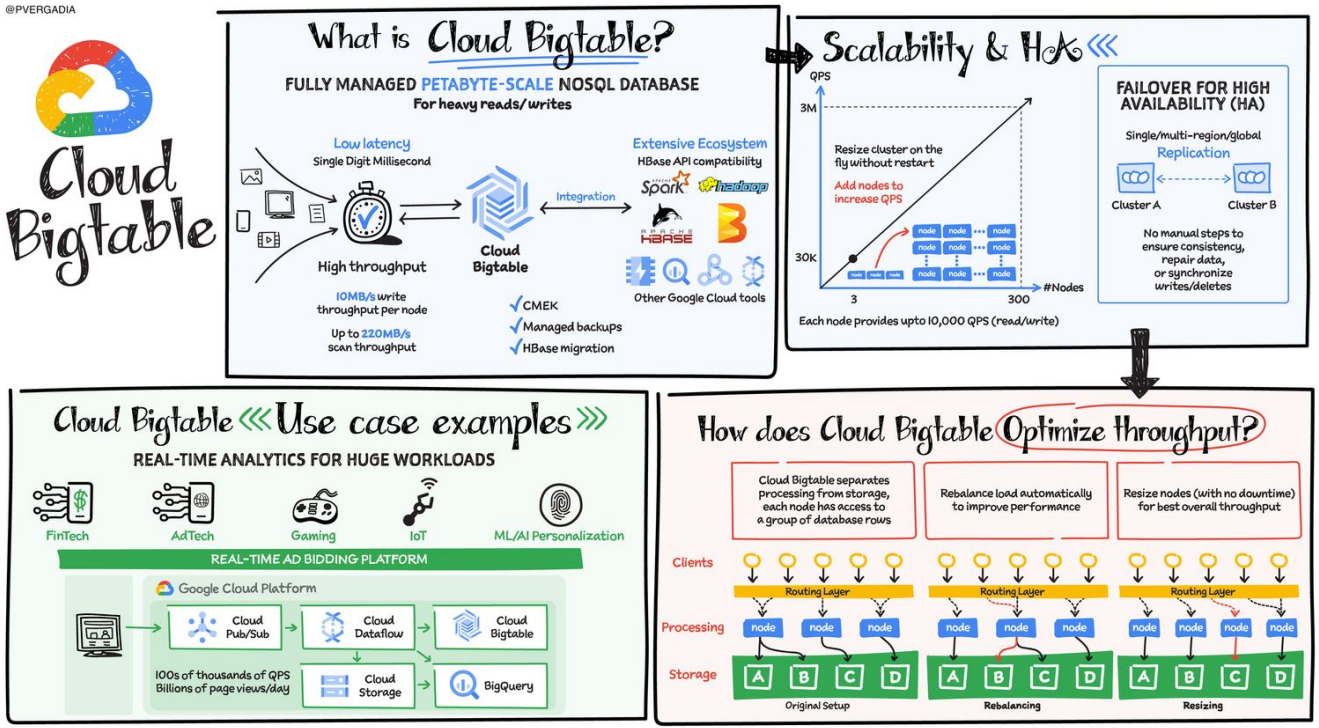
Big Table



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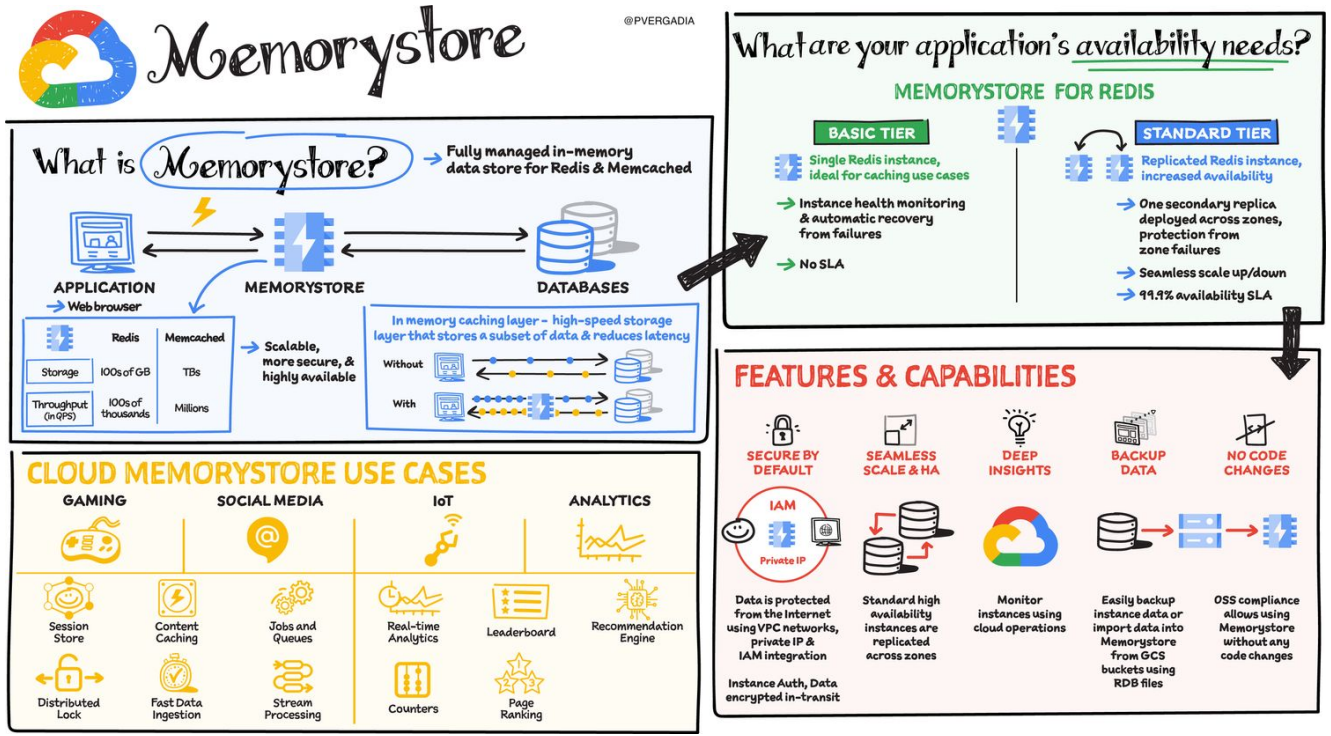
Memorystore



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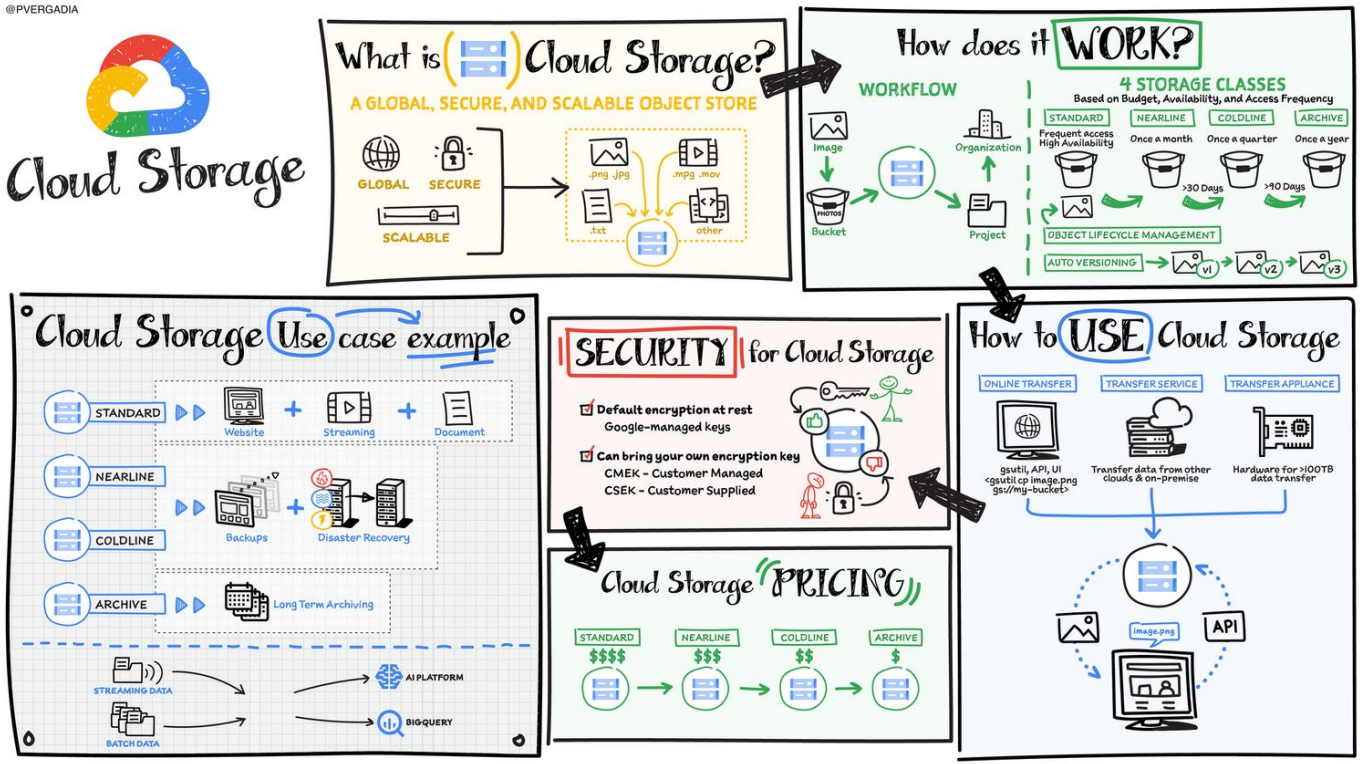
Cloud Storage



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Vector Database

Definition: A vector database is any database that allows you to store, index, and query *vector embeddings,* or numerical representations of unstructured data, such as text, images, or audio.

Vector embeddings are useful representations of unstructured data because they map content in such a way that semantic similarity is represented by distance in n-dimensional vector space. This makes it easy to search for similarity, find relevant content in a knowledge base, or retrieve an item that best matches a complex user-generated query.

Ranking solution dashboard

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Vertex AI Vector Search

Definition: Vector Search is based on vector search technology developed by Google research. With Vector Search you can leverage the same infrastructure that provides a foundation for Google products such as Google Search, YouTube, and Play.

Vector embeddings are useful representations of unstructured data because they map content in such a way that semantic similarity is represented by distance in n-dimensional vector space. This makes it easy to search for similarity, find relevant content in a knowledge base, or retrieve an item that best matches a complex user-generated query.

Overview of Vertex AI Vector Search

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PG Vector to be used with Cloud SQL

Definition: Open-source vector similarity search for Postgres. PG Vector is an extension of PostgreSQL that allows to store embedding of vectors.

Cloud SQL is the GCP product allowing to create PostgreSQL database. The PG Vector extension can be added to the database.

Tutorial to use PG Vector in Cloud SQL

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Question 1

You are designing a database architecture for a global application that stores information about public parks worldwide. The application uses the database for read-only purposes, and a centralized batch job updates the database nightly. You want to select an open source, SQL-compliant database. What should you do?

A. Use Bigtable with multi-region clusters.

B. Use Memorystore for Redis with multi-zones within a region.

C. Use Cloud SQL for PostgreSQL with cross-region replicas.

D. Use Cloud Spanner with multi-region configuration.

Answer explanation

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Question 2 - 3

You have a very large database that you're using for complex queries in a suite of business intelligence applications. You want to move the data to a fully-managed solution. Which option is ideal for such use cases?

- Cloud Bigtable
- Firestore
- BigQuery
- Cloud SQL

You're building a banking application that is expected to have a very large number of users around the world. When users make a deposit, they want to see the result of this deposit reflected immediately when they view their balance. What data storage option is ideal for storing account balance information for users?

- Cloud Bigtable is ideal because it supports low-latency read/write access.
- Firestore is ideal because it lets you develop a mobile app.
- Cloud Spanner is ideal because it supports strongly consistent reads in addition to horizontal scalability, low latency, and high throughput.
- Cloud SQL is ideal because it's a relational database that supports transactions.

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Question 4

You are building an application that stores relational data from users. Users across the globe will use this application. Your CTO is concerned about the scaling requirements because the size of the user base is unknown. You need to implement a database solution that can scale with your user growth with minimum configuration changes. Which storage solution should you use?

A. Cloud SQL

B. Cloud Spanner

C. Cloud Firestore

D. Cloud Datastore

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Question 5

Your organization has a ticketing system that needs an online marketing analytics and reporting application. You need to select a relational database that can manage hundreds of terabytes of data to support this new application. Which database should you use?

A. Cloud SQL

B. BigQuery

C. Cloud Spanner

D. Bigtable

Answer explanation

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Our use case

- We have files to store for ou RAG chatbot, where to store raw files, images, videos, audios … ?
- Once these files are stored, we chunk, convert into embeddings. Where do we store them ?
- When user gives a feedback in ou interface, where do we store them for analytics ?
- If we had a username/password to our interface, where to we store them ?

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Conclusion

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The project

- How to create a RAG / Agent architecture ?
- How and where to store our Data ?
- How to create a frontend application in Python (Streamlit, Gradio, Chainlit) ?
- How to create a Docker image containing the Python code ?
- How to deploy an application to access the frontend using internet ?
- How to secure the application ?
- How to add multimodal data processing ?
- How to get feedback from the user ?
- Where to store the feedback ?
- How to create a feedback dashboard ?
- How to evaluate our chatbot ?

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